

# HPC exam project - part 3

A.Y. 2025-26

This exercise extends the previous ones on simulating damped wave motion using OpenMP and MPI. In the third version of the exercise, the goal is to color the resulting image to highlight the maximum and minimum values of the simulated waves while smoothing (fading) the wave fronts. In addition, to make the simulation more visually clear, it is suggested that the regions of the matrices representing calm areas, or areas near to zero, have to be colored with a clear color such as white.

It is possible to render the simulation video in color by saving each frame in PPM format, a generalization of the previously used PGM grayscale format. In particular, in the PPM format, each pixel is represented by a triplet of numbers ranging from 0 to 255; each number represents the intensity of a color component in the RGB (RED, GREEN, BLUE) representation. For example, pure red is represented as (255,0,0), black as (0,0,0), and white as (255,255,255).

A PPM file has the same structure as a PGM file, but the number of rows and columns must be interpreted as the number of RGB value triplets present. For example, a 10×10 image contains 100 pixels, corresponding to 300 total numerical values; these values are organized in a 10×30 table. Furthermore, the P3 encoding is used, where each value is written in ASCII text format.

The following matrix represents a 5×5 pixel surface that is entirely green with a diagonal line of black pixels:

|         |         |         |         |         |
|---------|---------|---------|---------|---------|
| 0 255 0 | 0 255 0 | 0 255 0 | 0 255 0 | 0 0 0   |
| 0 255 0 | 0 255 0 | 0 255 0 | 0 0 0   | 0 255 0 |
| 0 255 0 | 0 255 0 | 0 0 0   | 0 255 0 | 0 255 0 |
| 0 255 0 | 0 0 0   | 0 255 0 | 0 255 0 | 0 255 0 |
| 0 0 0   | 0 255 0 | 0 255 0 | 0 255 0 | 0 255 0 |

Using CUDA, it is possible to execute the code on the GPU and transform each point in the grayscale matrix into a properly represented RGB triplet. A suitable mathematical transformation function must be implemented to apply colors to the simulation; in particular, it is necessary to clearly identify the maximum and minimum points in the wave with very bright colors. Consider to color the wave slopes (fall and rising fronts) with different colors, gradually fading the shades from the maximum to the minimum points. The recolored matrix is then saved to a file as **frame\_<xxxxx>.ppm**, while the original matrix is used to continue the simulation.